

is immediately filled by a B–D packet, and the throughput of the A–C connection at R2 goes to zero. This, in turn, *implies that the A–C end-to-end throughput goes to zero* in the limit of heavy traffic. These considerations give rise to the offered load versus throughput tradeoff shown in Figure 3.48.

The reason for the eventual decrease in throughput with increasing offered load is evident when one considers the amount of wasted work done by the network. In the high-traffic scenario outlined above, whenever a packet is dropped at a second-hop router, the work done by the first-hop router in forwarding a packet to the second-hop router ends up being “wasted.” The network would have been equally well off (more accurately, equally bad off) if the first router had simply discarded that packet and remained idle. More to the point, the transmission capacity used at the first router to forward the packet to the second router could have been much more profitably used to transmit a different packet. (For example, when selecting a packet for transmission, it might be better for a router to give priority to packets that have already traversed some number of upstream routers.) *So here we see yet another cost of dropping a packet due to congestion—when a packet is dropped along a path, the transmission capacity that was used at each of the upstream links to forward that packet to the point at which it is dropped ends up having been wasted.*

3.6.2 Approaches to Congestion Control

In Section 3.7, we’ll examine TCP’s specific approach to congestion control in great detail. Here, we identify the two broad approaches to congestion control that are taken in practice and discuss specific network architectures and congestion-control protocols embodying these approaches.

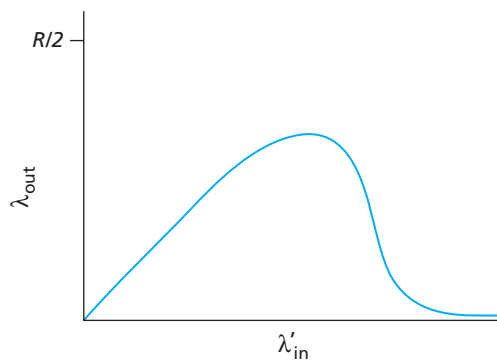


Figure 3.48 ♦ Scenario 3 performance with finite buffers and multihop paths

At the broadest level, we can distinguish among congestion-control approaches by whether the network layer provides any explicit assistance to the transport layer for congestion-control purposes:

- *End-to-end congestion control.* In an end-to-end approach to congestion control, the network layer provides *no explicit support* to the transport layer for congestion-control purposes. Even the presence of congestion in the network must be inferred by the end systems based only on observed network behavior (for example, packet loss and delay). We will see in Section 3.7 that TCP must necessarily take this end-to-end approach toward congestion control, since the IP layer provides no feedback to the end systems regarding network congestion. TCP segment loss (as indicated by a timeout or a triple duplicate acknowledgment) is taken as an indication of network congestion and TCP decreases its window size accordingly. We will also see a more recent proposal for TCP congestion control that uses increasing round-trip delay values as indicators of increased network congestion.
- *Network-assisted congestion control.* With network-assisted congestion control, network-layer components (that is, routers) provide explicit feedback to the sender regarding the congestion state in the network. This feedback may be as simple as a single bit indicating congestion at a link. This approach was taken in the early IBM SNA [Schwartz 1982] and DEC DECnet [Jain 1989; Ramakrishnan 1990] architectures, was recently proposed for TCP/IP networks [Floyd TCP 1994; RFC 3168], and is used in ATM available bit-rate (ABR) congestion control as well, as discussed below. More sophisticated network feedback is also possible. For example, one form of ATM ABR congestion control that we will study shortly allows a router to inform the sender explicitly of the transmission rate it (the router) can support on an outgoing link. The XCP protocol [Katabi 2002] provides router-computed feedback to each source, carried in the packet header, regarding how that source should increase or decrease its transmission rate.

For network-assisted congestion control, congestion information is typically fed back from the network to the sender in one of two ways, as shown in Figure 3.49. Direct feedback may be sent from a network router to the sender. This form of notification typically takes the form of a **choke packet** (essentially saying, “I’m congested!”). The second form of notification occurs when a router marks/updates a field in a packet flowing from sender to receiver to indicate congestion. Upon receipt of a marked packet, the receiver then notifies the sender of the congestion indication. Note that this latter form of notification takes at least a full round-trip time.

3.6.3 Network-Assisted Congestion-Control Example: ATM ABR Congestion Control

We conclude this section with a brief case study of the congestion-control algorithm in ATM ABR—a protocol that takes a network-assisted approach toward congestion control. We stress that our goal here is not to describe aspects of the ATM architecture